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Key Personnel Changes/Previous PI:			
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Task Description:	Introduction. Spaceflight exposes crewmembers to sleep loss and circadian misalignment, which impair cognition and increase the risk of errors and accidents. Light has both circadian resetting and direct alerting effects, therefore, we tested the effects of an experimental dynamic lighting schedule (DLS) on circadian phase, sleep, and performance during a 45-day simulated space mission. Methods. Twenty participants [mean age (±SD) 38.7 ± 8.2 years; 7F] were studied in 4-person teams at the NASA Human Exploration Research Analog (HERA) Campaign 4 at the Johnson Space Center (JSC). Participants were scheduled to sleep for six 7-day cycles of 8 h/night for two consecutive nights then 5 h/night for 5 nights, ending with three 8-h recovery sleep episodes, aligned by waketime. We conducted a randomized between-subject trial to compare the effects of a dynamic lighting schedule (DLS) (n=8) with a static standard lighting schedule (SLS) (n=8). One mission (n=4) was excluded for light compliance failure. In the DLS, crewmembers were exposed to blue-enriched white light during all wake episodes (6500K) except for 3 hours before bedtime when the lighting was reduced in intensity and short-wavelength content (4100K, 2700K). Lighting was constant for all wake episodes in the standard condition. Circadian phase was calculated weekly from urinary 6-sulphatoxymelatonin (aMT6s) acrophase time, sleep was assessed using daily sleep logs completed before and during the mission, and cognition was assessed 3 times per week throughout the 45-day mission. Results. The average (±SE) pre-mission aMT6s acrophases did not differ between the DLS (3.69 ± 0.38 h, n=8) and SLS (4.24 ± 0.38 h, n=8) conditions. The average acrophase was 1.55 ± 0.13 h later in the SLS compared to the DLS across the 45-day mission ($p<0.05$), with the largest difference being 1.79 ± 0.13 h on MD 33. The variability in acrophases across the 45-day mission was also significantly higher in the SLS (mean ± SD: 0.62 ± 0.06 h) than in the DLS condition (0.47 ± 0.04 h, p=0.04). The mean sleep duration on weeknights was significantly longer in the DLS (5.08 ± 0.04 h) compared to the SLS condition (4.95 ± 0.05 h, p=0.02), but similar between the conditions on weekend nights (7.50 ± 0.05 h vs. 7.30 ± 0.05 h, respectively, p=0.41). There was no overall difference in sleep onset latency (SOL) between conditions. Among the 10 cognitive tests, accuracy on the Abstract Matching and Matrix Reasoning Task (MRT) was significantly better in the DLS condition compared to SLS condition [p=0.020 and p=0.004 (FDR-adjusted p=0.039 and p=0.008), respectively]. Discussion. Compared to the SLS, the DLS significantly alleviated the drift in circadian phase typically observed in isolation analog studies, reduced the prevalence of sleep episodes occurring at an adverse circadian phase, and improved several mission-critical cognitive domains. Our results support incorporating DLS in future missions, which may facilitate appropriate circadian alignment and minimize the risk of sleep- and circadian-related decrements in performance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The Earth-based application and commercialization potential of these studies are enormous. Electric lighting is ubiquitous and can improve alertness directly, thereby improving productivity and safety, and ultimately health. Anywhere where electric lighting is used, including schools, colleges, offices, factories, hospitals, care homes, residential homes, and transport and military applications, are potential beneficiaries. The International Space Station (ISS) ‘slam-shifts’ and the consequences to sleep, circadian rhythms, and performance, are very similar to that experienced by 15M American shiftworkers. Similar lighting interventions could be deployed for night shift-workers to improve productivity and safety while also reducing energy use. Many non shift-workers also suffer circadian misalignment, poor daytime alertness, and performance problems, and would also benefit from a dynamic lighting schedule.
	In a series of 45-day 2016 HERA Campaign missions, we will conduct randomized between-subject clinical trials to test the hypotheses that deployment of the Dynamic Lighting Schedule (DLS) (n=12), as compared to deployment of a static Standard Lighting Schedule (SLS) (n=8) will: i) significantly improve circadian rhythm stability adaptation during the mission, as measured using the circadian rhythm of urinary 6-sulfatoxymelatonin ; ii) significantly improve subjective and actigraphic measures of sleep latency, sleep duration, sleep quality, and sleep efficiency; iii) significantly improve cognitive performance and subjective alertness (COGNITION battery). METHODS Participants. The HERA Campaign anticipated 4 missions per year (one per quarter) with 4 crew members per mission, and therefore we originally proposed to study 16 subjects over the 12-month duration of this award. Starting in 2017, the missions were scheduled for 45-day durations. Missions were planned to start in February 2017, but the initial mission in our series did not start until May 2017. In addition, the second mission in this four-mission series was aborted on day 23 due to adverse weather conditions. NASA therefore decided to add a fifth mission to this HERA campaign, which increased our overall sample to 20 subjects (mean age ± SD = 38.7 ± 8.2 years; 7 females). All crewmembers who participated were screened medically, psychologically, and behaviorally prior to the study. All recruitment, screening, and study oversight was provided by NASA personnel. Environment. HERA is a two-story, four-port cylindrical habitat unit at Johnson Space Center which includes a core, loft simulated airlock, and hygiene module. In addition to the physical simulation, HERA can also simulate communication methods in addition to ‘surveillance video and audio system, flight-like timeline and procedure viewer’ similar to space missions. Protocol. The 45-day protocol consists of 6 cycles of 2 night x 8 hours’ sleep (23:00-7:00) followed by 5 nights of 5 hours’ sleep, plus 3 recovery sleep episodes at the end, aligned by waketime. The DLS uses lights of three different spectral characteristics to address different operational needs: 1) white light for general vision; 2) blue-enriched high intensity white light to enhance alertness and circadian adaptation; 3) blue-depleted low intensity white light to minimize alertness prior to sleep.

Lighting. We made an initial review of lighting information made available to us and conducted additional on-site measurements to measure current lights and the proposed replacement lights in situ.

Standard Lighting Schedule (SLS): The general approach for the Standard Condition is to have fixed spectrum standard lamps with a nominal Correlated Color Temperature (CCT) at approximately 4000K, in the mid-range of possible melanopic Daylight Equivalent Ratio (DER) values, at ~0.5-0.7 and at medium intensity (while complying with visual lighting standards). The same lights will be used at night if needed. All of these lamps will be available off-the-shelf or are already installed in HERA.

Dynamic Lighting Schedule (DLS): The general approach for the Dynamic Condition is to use the following pattern each day: - Day: Blue-enriched medium intensity light (5500-6500K, melDER >0.7) ; - Starting 3 h pre-sleep: Blue-depleted medium intensity light (3000-4000K, melDER 0.5-0.6) ; - Starting 1 h pre-sleep: Blue-depleted low intensity light (2500-3500K, melDER 0.4-0.6) ; - Night: Blue-depleted very low intensity light (2500, melDER 0.4)

New standard international (SI) units have been provided by the International Commission on Illumination (CIE) to define light for these purposes (CIE, 2018) including the melanopic Equivalent Daylight (D65) Illuminance (EDI), a measure of light quantity stimulating the non-visual photoreceptor system, and melanopic Daylight Equivalent Ratio (DER), which provides an overall indication of the relative melanopic strength of the spectra; higher ratios approaching or greater than 1 indicate an strong stimulus strength, whereas values <0.5 are weaker.

It was decided by NASA that the order of light intervention would be Standard Condition for the first two missions followed by Dynamic Condition for the last two missions. The fifth mission was also scheduled under the Dynamic Condition.

OUTCOMES: Circadian phase was assessed using urinary 6-sulphatoxymelatonin (aMT6s). Participants collected 4-hourly urine samples (5-8 hours overnight) for 48 hours twice pre-mission (starting 10 and 5 days before mission) and mid-week during each week of the mission (starting on mission days 5, 12, 19, 26, 33, 43). Daily sleep diaries were completed by each participant every morning, post-sleep, including for 14 days pre-mission. Cognitive performance was assessed using the Cognition test battery for spaceflight which was administered every other day before the mission (MD-1) at ~19:00 on days with 5 hours of sleep. Participants completed a weekly Survey asking crewmembers to rate the severity of several symptoms related to physical and emotional well-being, daytime sleepiness, and the overall acceptability of the lighting environment on work and other activities of daily living.

RESULTS

Lighting. On average, the visual (photopic) illuminance at 72” in the horizontal plane on Level 1 increased from 467 to 1210 lux during the day in the DLS but reduced to 44 lux in the evening. The DLS increased melanopic EDI nearly 4-fold during the day and reduced it by over 90% in the evening. The melanopic DER changed from 0.71 to 0.91 during the day and from 0.71 to 0.48 in the evening between the SLS and DLS conditions, respectively. On Level 2, the photopic lux was similar during the day for both conditions (108 vs 125 lux) but reduced from 124 lux to 5 lux or less in the evening during the DLS, with similar changes in the melanopic EDI. The DER value for Level 2 increased from 0.58 for the SLS to 0.69 for the DLS during the day but was similar in the evening (0.55-0.59). The major change in the bunk lighting was a reduction in intensity by 4-5 fold (including for melanopic EDI) but there was no change in the light spectrum.

Task Progress:

Circadian phase. The average (±SE) pre-mission aMT6s acrophases did not differ between the DLS and SLS conditions ($p > 0.05$). The mean acrophase was 1.6 hours later in the SLS compared to the DLS across the 45-day mission ($p < 0.05$). There was no interaction between lighting condition and mission day on acrophase times. The variability in acrophases across the 45-day mission was significantly higher in the SLS than in the DLS condition ($p<0.05$). Amongst the 83 acrophases estimated during the mission, the proportion of acrophases that were optimal (i.e., between 2:45 am and 5:39 am [(± 1.0 SD for the mean in sighted individuals]) was higher in the DLS condition (45/48) compared to the SLS condition (13/35). The odds of optimal urinary aMT6s acrophase was 3.5 times higher in the DLS compared to SLS condition [RR: 3.46 (95% CI: 1.43 – 8.36), p=0.002].

Sleep. On average, sleep duration on weeknights (with 5 hours of time-in-bed scheduled) in the DLS condition (5.08 ± 0.04 h) was significantly longer than in the SLS condition (4.95 ± 0.05 h, p=0.02). In contrast, sleep duration on weekend nights (8 hours of time-in-bed scheduled) was not different between DLS and SLS conditions (7.50 ± 0.05 h vs. 7.30 h ± 0.05 h, respectively, p=0.41). There was no overall difference in sleep onset latency (SOL) between conditions. Amongst the 556 sleep episodes analysed, the proportion of sleep episodes that were aligned with optimal aMT6s acrophase (i.e., aMT6s acrophase occurred during scheduled sleep and between 2:45 – 5:39 am for that calendar day) was higher in the DLS condition (93%, 325/348) compared to the SLS condition (43%, 89/208). The odds of having optimally aligned sleep was significantly higher in the DLS condition compared to the SLS condition [(RR: 5.17, 95% CI 1.11-24.13), p=0.033]. Furthermore, when analysed by lighting condition, sleep duration was significantly longer in the DLS condition compared to the SLS condition when sleep was aligned with optimal urinary aMT6s acrophase on weekend nights (p=0.03) but not on weekday nights (p=0.39). When sleep was misaligned with optimal urinary aMT6s acrophase, sleep duration was significantly longer on weeknights (p=0.01) but not on weekend nights (p=0.07). SOL was significantly shorter under DSL compared to SLS on both weekend and week nights in the misaligned state ($p<0.0001$) but not when aligned.

Cognitive performance. Among the 10 cognitive tests, reaction time on the Motor Praxis (MP) Test trended to be faster in the DLS compared to the SLS (p=0.033, FDR-adjusted p=0.066), but otherwise no statistically significant differences were observed between the lighting conditions for any of the other tests. Accuracy on the Abstract Matching and Matrix Reasoning Task (MRT) was significantly better in the DLS condition compared to SLS condition [p=0.020 and p=0.004 (FDR-adjusted p=0.039 and p=0.008), respectively], and trended to be better in the MP test in the DLS compared to the SLS (p=0.078, FDR-adjusted p=0.078), but otherwise no statistically significant differences were observed between the lighting conditions for any of the other tests.

Surveys. There were no significant differences between the missions in terms of physical or mental symptoms related to the sleep schedule or the lighting environment.

DISCUSSION

The HERA Campaign is considered a high-fidelity analog of space missions, mimicking the isolation and communication delays, team dynamics under stressful conditions, high workloads, and schedules typical of space flight.

	While sleep has often been assessed during these missions, circadian phase assessment using validated measures has rarely been performed. In the current study, we assessed the timing of urinary 6-sulphatoxymelatonin (aMT6s), a gold-standard marker of circadian phase for field studies, each week. The DLS was successful at improving circadian entrainment. Under the SLS, the average shift in the timing of the aMT6s peak was 1.6 h greater than under DLS. While these absolute differences in phase do not appear substantial, the differences are remarkable in that they occurred under highly restricted conditions with the same photoperiod in both groups, i.e., the timing of when it was light or dark was not changed, only the properties of light during the scheduled wake episodes. Circadian phase varied less under the DLS and therefore could be considered more strongly entrained to the light-dark cycle. We did not observe major changes in sleep in the current mission. Under the chronic variable sleep schedule imposed, crewmembers maximized the sleep opportunities and fell asleep quickly, limiting the degree to which sleep duration and latency could differ between lighting conditions. There was evidence of reduced sleep duration and longer sleep latency during the SLS condition, particularly when misaligned, highlighting the utility of measuring circadian phase for interpretation of behavior. We also did not observe major effects of the light intervention on performance, measured once in the evening on selected days. Only the Abstract Matching (AM) and Matrix Reasoning (MRT) tests showed a showed a modest improvement in accuracy during the DLS overall. The standard lighting data demonstrate why appropriate lighting countermeasures should be incorporated proactively in every space mission and spaceflight analog in order to avoid the predictable risk of circadian misalignment, and its subsequent impact on sleep, performance, and safety. To this end, NASA has embarked on a lighting replacement program for ISS that exchanges the older General Luminaire Assemblies (GLA) that contain fluorescent lamps with a Solid State Lighting Assembly (SSLA) containing programmable Light Emitting Diode (LED) lamps. We (Brainard and Lockley laboratories) are currently assessing the impact of this new lighting on visual function, circadian phase, sleep, and cognition. In summary, the DLS led to more stable entrainment of the circadian system, leading to a reduction in the number of sleep episodes occurring at an adverse circadian time and subsequently better sleep during the weekday restricted sleep nights. Multiple missions, including the current one, demonstrate that simply relying on standard lighting is insufficient to maintain appropriate circadian entrainment and results in impaired sleep and performance. Reference: CIE. S 026/E:2018. CIE System for Metrology of Optical Radiation for ipRGC-Influenced Responses to Light. https://
Bibliography Type:	Description: (Last Updated: 11/14/2025)
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